

Fair-Fracture: A YOLOv8 and U-Net Framework for Bone Fracture Diagnosis

1st Agnim Gupta

*Department of Computer Science
Stevens Institute of Technology
Hoboken, New Jersey, U.S.A
agupta57@stevens.edu*

2nd Jay Gajjar

*Department of Mathematical Science
Stevens Institute of Technology
Hoboken, New Jersey, U.S.A
jgajjar@stevens.edu*

3rd Tejas Nidhankar

*Department of Electrical and Computer Engineering
Stevens Institute of Technology
Hoboken, New Jersey, U.S.A
tnidhank@stevens.edu*

Abstract—Bone fracture identification via X-ray imaging is important in emergency medicine, where timely detection impacts patient outcomes. Manual inspection is prone to errors when radiologists are overworked. This research presents Fair-Fracture, a hybrid deep learning system that combines U-Net semantic segmentation with YOLOv8 object detection for comprehensive fracture diagnosis. Our pipeline uses two complementary datasets: FracAtlas for pixel-level U-Net segmentation training, and a custom dataset from Roboflow spanning 17 fracture classes for YOLOv8 multi-class detection. U-net model achieved approximately 99.26% segmentation accuracy and a Dice score of around 0.30, reliably identifying fracture regions at the pixel level. YOLOv8 achieved an overall mAP50 of 0.364 across 17 fracture types, with strong performance on well-represented classes such as closed-simple fractures (mAP50 = 0.891). By deploying a U-Net model for localized pixel-level segmentation alongside a YOLOv8 model for multi-class type identification, our approach provides a comprehensive analysis, offering clinicians both high-precision fracture mapping and specific categorical diagnosis

Index Terms—bone fracture detection, YOLOv8, U-Net, semantic segmentation, object detection, medical image analysis

I. INTRODUCTION

According to the World Health Organization, bone fractures are one of the most common injuries seen in emergency rooms around the world, accounting for an estimated 178 million new cases each year [16]. Identification of fractures needs to be fast and accurate to optimize treatment planning and patient outcomes. The current standard of care is a manual review of X-rays by trained radiologists. It works, but it is time-consuming and error-prone, especially when dealing with many patients or subtle fracture patterns.

Recent advances in deep learning have shown their potential to transform the analysis of medical images. In diagnostic tasks such as screening for diabetic retinopathy [6], classification of skin cancer [4] and detection of pneumonia on chest radiographs [12], Convolutional Neural Networks (CNNs) have demonstrated expertise-level performance. Deep learning may enable automated screening for bone fracture detection,

which may reduce diagnosis time, reduce missed fractures, and aid radiologists in decision-making in high-paced settings.

However, existing fracture detection methods typically focus on binary classification (fractured or not fractured), yet rarely provide the simultaneous pixel-level localization and multi-class identification required to distinguish between specific fracture types. Moreover, much of the prior research concentrated on binary classification of simple fracture patterns. The complex classification of fracture types, such as comminuted, oblique, spiral, and stress fractures, has been relatively under-explored in automated diagnostic systems.

To overcome these limitations, we present a novel architecture that leverages two specialized deep learning models in parallel. YOLOv8 can identify 17 different kinds of fractures. In addition to classifying the fracture, it can use bounding boxes to approximate its location. Binary segmentation is performed using U-net to precisely shows the extent of fractured regions at the pixel level. Fracture boundaries and morphology can be seen in detail thanks to it. To provide medical professionals with a comprehensive diagnosis, our Fair-Fracture framework runs two models in parallel and displays their outputs: the type of fracture (YOLOv8 classification), its location (U-Net segmentation).

Approach	Outputs Provided
CNN classifier only	Class label
Object detector (YOLO)	Class + bounding box
Segmentation	Pixel-level mask
Fair-Fracture (ours)	Class + box + pixel mask

This paper’s primary contributions are:

(1) a dual-model pipeline that uses U-Net semantic segmentation and YOLOv8 object detection in parallel for thorough fracture analysis, providing classification, localization, and pixel-level extent in a single pass;

(2) Two complementary datasets were used for evaluation: a custom dataset from Roboflow for multi-class detection with

2,000 images covering 17 types of fractures, and FracAtlas for binary segmentation training with 4,083 images annotated by expert radiologists;

(3) a thorough performance analysis across fracture classes with different levels of detection difficulty demonstrates how the availability of training data and visual distinctiveness affects model performance; and

(4) A clinical need-driven design with an emphasis on actionable outputs for emergency department workflows, interpretability (confidence scores), and computational efficiency.

II. RELATED WORK

A. Deep Learning in Medical Image Analysis

The use of deep learning in medical imaging has grown rapidly since the introduction of AlexNet [9], which demonstrated that convolutional neural networks could learn hierarchical feature representations directly from raw pixel data. This eliminated the need for handcrafted features common in older computer vision methods. This was especially useful in medical imaging, where the visual patterns important for diagnosis are often subtle and hard to produce by hand.

Specifically for bone fracture detection, Chung et al [2] showed that deep CNNs could detect wrist fractures on traditional radiographs just as well as, or better than, radiologists. Convolutional architectures may efficiently extract discriminative features from X-ray images without requiring bone segmentation or anatomical landmark recognition, according to their research.

B. Object Detection Architectures

The YOLO architecture family [13] transformed how we detect objects by treating detection as a single regression problem and predicting bounding boxes and class probabilities directly from the full image in a single pass through the network. This is different from earlier two-stage detectors like R-CNN [5], which first generate region proposals and then classify each one independently. The single-stage method is much faster, which makes it a good choice for real-time clinical use.

The latest version of the family, YOLOv8, has several architectural improvements, such as a more efficient feature pyramid neck, an anchor-free detection head, and an improved CSPDarknet backbone with cross-stage partial connections. These changes lead to better feature extraction and more accurate bounding-box regression. This is especially important for detecting small fracture lines in X-ray images, which are often small relative to the overall image.

C. Semantic Segmentation in Medical Imaging

U-Net [14] introduced a skip-connected encoder-decoder architecture, which is now the industry standard for medical image segmentation. The symmetric expanding path combines the high-resolution features from the contracting path with the upsampled features from the previous layers for precise localization. For fracture analysis, pixel-level segmentation is preferable to bounding boxes, as it clearly delineates the

affected areas and shows the exact extent of the fracture, which is helpful for surgical planning. For CT scans of bone fractures, Zhou et al. [17] showed U-Net variants may achieve Dice coefficients over 0.90.

D. Vanishing Gradients and Deep Architectures

The vanishing gradient problem happens when the depth of a neural network increases, and gradients fade exponentially as they move through many layers during backpropagation [8]. He et al. [7] utilized residual learning, a form of skip connection that allows gradients to traverse directly through the identity mappings, to address this problem. This lets you train networks with hundreds of layers.

E. Dual Model Approaches for Fracture Detection

Son et al. [15] trained a YOLOv4 object identification module and a U-Net segmentation module separately to diagnose mandibular (Jawbone) fractures on panoramic radiographs. YOLOv4 was good at finding the bounding boxes but it was not able to find all of them due to the irregular shapes of the fractures. U-Net applied pixel segmentation to identify many missed cases and improved the overall recall from 79% to 87% at 95% precision. This was used to diagnose fractures by combining different detection and segmentation models. Our study builds on this and classifies 17 different types of musculoskeletal fractures using the more advanced YOLOv8 architecture.

III. OUR SOLUTION

Our solution utilizes a dual-model deep learning pipeline that independently deploys two complementary architectures for comprehensive bone fracture diagnosis from a single X-Ray: a U-Net specifically for pixel-level segmentation and a YOLOv8 detector for multi-class fracture classification. This section describes the datasets used, the machine learning algorithms selected, our implementation details, and the novel contribution of our work. Fig. 1 presents an overview of the parallel training and inference pipeline.

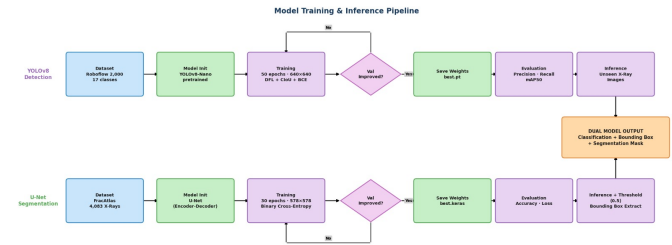


Fig. 1. Parallel training and inference pipeline for YOLOv8 and U-Net, merging into a hybrid output.

A. Description of the Dataset

Our pipeline has two complementary datasets for the model's two branches. U-net classification branch is trained on the FracAtlas [1] dataset, a publicly available collection

of musculoskeletal radiographs comprising 4,083 anonymized images from major hospitals in Bangladesh. The dataset is annotated by two expert radiologists and validated by an orthopedic surgeon, providing pixel-level polygon masks of fracture regions in the hand, leg, hip, and shoulder. Annotations are stored in VGG JSON format produced by the VGG Image Annotator tool; each fracture region is encoded as a list of polygon vertex coordinates. Exploratory data analysis revealed that the dataset was imbalanced, with 3,366 non-fractured samples (82.4%) and 717 fractured samples (17.6%), reflecting realistic clinical prevalence ratios. Only images from the Fractured subset with corresponding polygon annotations are used for segmentation training.

YOLOv8’s detection branch is trained on the Bone Fracture dataset available on Roboflow Universe [15], which contains 2000 annotated X-ray images of 17 different fracture classes, including avulsion, closed-simple, comminuted, compression-crush, fracture-dislocation, greenstick, hair-line, impacted, intra-articular, longitudinal, oblique, open-compound, pathological, segmental, spiral, stress, and transverse. Dataset is provided in native YOLOv8 format, with a data.yaml config file with the class names and paths to train/validation/test. Each image is labeled with one or more axis-aligned bounding boxes and class indices. This dataset offers the multi-class classification, which FracAtlas alone cannot provide.

When FracAtlas loads a dataset, it uses PIL’s ImageDraw function to turn raw polygon annotations which it gets from the VGG JSON file from the dataset into binary masks. Each polygon is filled with the value 255 (fracture), and the rest is left at 0 (background). Images are resized to 578×578 pixels using InterpolationMode.BILINEAR (to keep smooth intensity gradients), and masks are resized using interpolation InterpolationMode.NEAREST (nearest-neighbor) to preserve the binary 0, 255 domain, since bilinear interpolation would introduce fractional pixel values that corrupt the label. A seeded random generator is used to divide the 4,083 images into three groups: training (70%), validation (15%), and test (15%). To effectively manage all this data processing, PyTorch’s Dataloaders were implemented. The Ultralytics framework automatically preprocesses the Roboflow dataset. It letterboxes images to 640×640 while keeping the aspect ratio, and it applies training-time augmentations (mosaic, random horizontal flip, HSV perturbation, random translation, and scaling) on the fly to make the model more robust to changes in X-ray exposure and patient positioning.

B. Machine Learning Algorithms

1) *U-Net for dividing things up*: U-Net [10] is the most common approach for segmenting medical images. It has an encoder-decoder structure with skip connections that lets the model combine high-level semantic context (from deeper layers) with fine-grained spatial detail (from shallower layers). This is important for accurately defining the edge of a fracture at the pixel level. U-Net is excellent for medical imaging, as skip connections maintain high spatial resolution during

downsampling, and the design performs effectively with the limited datasets typical in medical contexts.

2) *YOLOv8 for Detection*: YOLOv8 [9] is a one-stage object detector that predicts bounding boxes and class labels in one forward pass. This makes it much faster than two-stage detectors like Faster R-CNN. This speed advantage is very important for clinical screening applications, where the amount of work that can be done in each time is critical. YOLOv8 also added an anchor-free detection head and separated the classification and regression branches, making it easier to detect small objects. This is important for finding subtle fracture lines. We picked YOLOv8-Nano (the smallest version, with about 3.2 million parameters) for our use case because it has a good balance between accuracy and latency and can run on regular hardware without a dedicated GPU.

TABLE I
MODEL ARCHITECTURE AND TRAINING SUMMARY

Property	U-Net	YOLOv8-Nano
Framework	PyTorch	Ultralytics
Input size	578 × 578	640 × 640
Dataset	FracAtlas	Roboflow Bone Fx
Epochs	30	50
Optimizer	Adam (1e-4)	Ultralytics default
Loss	0.5·BCE + 0.5·Dice	Ultralytics default
Output	Pixel mask	Class + Bbox

3) *Why Both*: YOLOv8 can only classify fractures and give a rough bounding box. It cannot show the exact shape or size of the fracture. U-Net produces only pixel-level masks for binary fracture-vs-background detection; it can’t distinguish between different fracture types. Running both in parallel lets us obtain classification, bounding-box localization, and pixel-level extent in a single diagnostic pass. The two models also fail in different ways, so using both together reduces the chance of missing a diagnosis

C. Implementation Details

1) *U-Net Implementation*: U-net was developed from scratch in PyTorch. The encoder starts with a double-convolution block (3→64 channels) and then goes through four stages of down-sampling. In each stage, 2×2 max pooling and a double-convolution block are used. The channel depth goes from 64→128→256→512→1024. A double-convolution block is two layers that go one after the other (3×3 Conv → BatchNorm → ReLUs) with padding = 1 and batch normalization helping in stabilizing the learning process by standardizing the inputs through nominalization. The decoder is like an encoder, but it has four up-sampling stages that use 2×2 transposed convolutions, skip-connection concatenation with center-cropped encoder features, and double-convolution blocks. A final 1×1 convolution produces a single-channel logit map, and at inference, a sigmoid activation yields pixel-wise probability.

Training uses a combination of Binary Cross-Entropy and Dice Loss equally: $Loss = 0.5 \cdot Loss_{(BCE)} + 0.5 \cdot Loss_{(Dice)}$

Pure BCE performs poorly under heavy class imbalance, where fracture pixels typically represent less than 5% of the image, the model predicting "all background" achieves 95% pixel accuracy without learning anything. The Dice term directly optimizes region overlaps and is complementary to the pixel-wise BCE penalty. Loss is calculated for every batch and to accurately track the accumulated loss over each epoch this batch loss is multiplied by the current batch size (`image.size(0)`). The model is trained for 30 epochs with an Adam optimizer with a learning rate of 0.0001 and a batch size of 8. The checkpoint with the highest validation Dice score is selected to be the final model.

2) *YOLOv8 Implementation*: YOLOv8 training is done using the Ultralytics framework, which exposes a simple Python API. We initialize the model with pre-trained COCO weights (`yolov8n.pt`) to leverage transfer learning and fine-tune it on the Roboflow Bone Fracture dataset for 50 epochs at 640x640 resolution on an Apple M-series GPU (device = mps). The default composite loss of the framework is a weighted sum of the Distribution Focal Loss (DFL) for learning a discrete distribution over box coordinates, Complete IoU (CIoU) loss for box geometry (overlap, center distance, and aspect ratio), and Binary Cross-Entropy for per-class classification. best is the best epoch weights (by val mAP).pt for inference. .

3) *Inference Pipeline*: At inference time, both models run independently on the same X-Ray. YOLOv8 produces one or more (class, bounding box, confidence) triplets. U-Net generates a pixel-wise probability map, which is thresholded at 0.30 to produce a binary mask. The threshold is set lower than usual 0.5 to increase sensitivity, which is appropriate for a screening setting in which missed fractures carry a high cost than false alarms. The mask is passed through OpenCV's `cv2.findContours` and contours with an area of less than 20 pixels are ignored as noise.

The resulting outputs are presented via a Gradio dashboard specifically designed to handle X-ray image samples that often contain multiple views of a limb within a single image. While YOLOv8 model analyzes the entire frame to list all identified fracture types in a dedicated "Detected Fracture Types" text box, the side-by-side visual panels allow the clinician to pinpoint exactly which view corresponds to which classification. By displaying the U-Net pixel-level mask and YOLOv8 bounding boxes simultaneously, the interface provides a visualization that connects the textual list of fracture types (example: spiral or oblique) to their specific locations across the various views in the X-ray

D. Novel Solution:

The novelty of our work is not in a particular model but in the parallel combination of a multi-class detector and a pixel-level segmentation model for bone fracture diagnosis. To the best of our knowledge, current work on automated fracture detection is limited to classification, object detection, or segmentation in isolation. The single-model approach is a compromise. Detectors can tell you the type of fracture and where it is, but not how big it is. Segmentation models can

tell you the size but not the type of fracture. Our pipeline runs YOLOv8 and U-Net models and their outputs provide fracture class label (from YOLOv8), location (bounding box from YOLOv8), and precise extent (pixel mask from U-Net) in a single diagnostic pass.

Another advantage of the parallel architecture is cross-verification: The bounding box from YOLOv8 and the bounding box obtained from U-net mask are produced by completely independent models trained on different datasets. There is strong evidence of a fracture where the two boxes overlap; where they disagree, the discrepancy is itself a useful signal that flags the case for expert review. This dual evidence design is directly motivated by the clinical workflow, where the AI system is meant to support the radiologist's judgment, not replace it.

IV. RESULTS

A. U-Net Segmentation Performance

U-net model was tested on the FracAtlas test set, which contained 15% of the 4,083 photos and was generated with a fixed seed so it could be reused. The model that used the combined BCE + Dice goal got a test pixel accuracy of 99.26% and a test loss of 0.3742. The training and validation loss curves in Figure 3 decreased steadily over 30 epochs. By epoch 20, they were nearly identical. This shows that the model generalized well. There was a small change after epoch 22, suggesting that minor overfitting was starting. Still, the checkpoint with the highest validation Dice score was kept because the validation Dice score continued to improve.

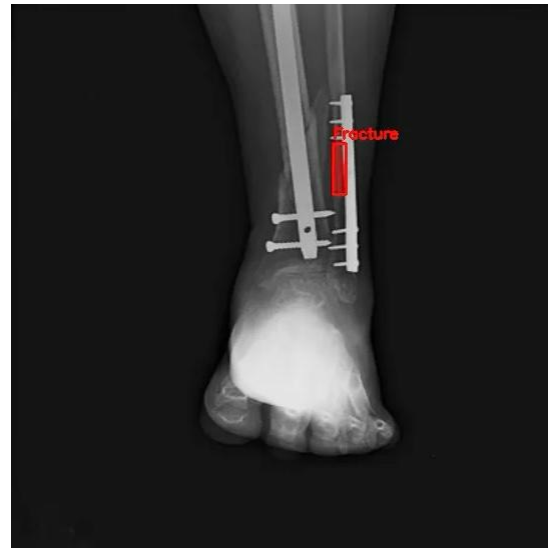


Fig. 2. U-Net inference output. A threshold binary mask with extracted bounding boxes overlaid on the original X-ray

It is important to remember that pixel accuracy alone may not work for segmentation when there is a big class imbalance. A degenerate model that predicts "all background" would already be more than 95% accurate without learning anything because fracture pixels make up less than 5% of an X-ray.

Using dice-based selection instead of accuracy-based selection to make sure that the stored model really finds the fracture sites. Figure 2 shows that U-net makes detection based on the clean mask that it generates.

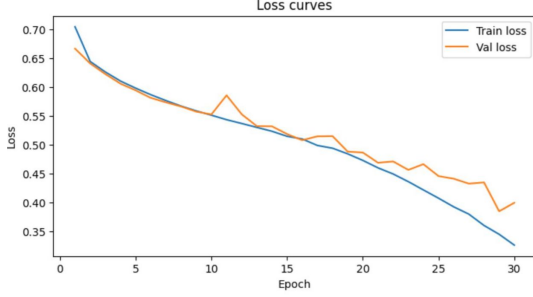


Fig. 3. U-Net training and validation loss curves over 30 epochs. Both curves decline together through epoch 20; mild divergence after epoch 22 indicates the onset of overfitting, but validation loss continues to improve.

B. YOLOv8 Detection Performance

YOLOv8 was evaluated on the Roboflow Bone Fracture validation set, comprising 82 photos and 91 fracture cases across 17 classes. The overall precision, recall, mAP50, and mAP50-95 of the model are 0.373, 0.423, 0.364, and 0.303, respectively. Here, mAP50 is the mean average precision in which the ground-truth box and the predicted bounding box overlap by at least 50% (a lenient criterion). mAP50-95: the average mAP over thresholds 50% to 95% with a step size of 5%. This tougher test requires closer alignment.

TABLE II
PERFORMANCE METRICS ACROSS FRACTURE CLASSES

Class	P	R	mAP50	mAP50-95
All (overall)	0.373	0.423	0.364	0.303
Closed-simple	0.640	0.909	0.891	0.732
Compression-crush	0.774	0.801	0.844	0.773
Stress	0.462	0.633	0.595	0.399
Open-compound	0.477	0.706	0.525	0.361
Greenstick	0.335	0.643	0.415	0.360
Comminuted	0.369	0.465	0.411	0.362
Fracture-dislocation	0.362	0.500	0.369	0.328
Avulsion	0.270	0.536	0.338	0.278
Hairline	0.207	0.321	0.313	0.270
Pathological	0.221	0.538	0.301	0.261
Intra-articular	0.262	0.277	0.277	0.253
Segmental	0.294	0.357	0.227	0.183
Impacted	0.186	0.110	0.169	0.157
Transverse	1.000	0.000	0.153	0.123
Spiral	0.247	0.263	0.139	0.126
Oblique	0.231	0.175	0.135	0.114
Longitudinal	0.000	0.000	0.081	0.074

As shown in Table III, performance varies widely across fracture types. The model performs well on closed-simple fractures (mAP50 = 0.891), compression-crush fractures (0.844),

and stress fractures (0.595), which are all well-represented classes with easy-to-see patterns. But classes like longitudinal (0.081), oblique (0.135), and transverse (0.153) have a large class imbalance (with only 7–18 training examples each). They are hard to find because these fractures can resemble overlapping anatomical structures or normal trabeculae.



Fig. 4. YOLOv8 normalized confusion matrix on the validation set. Diagonal entries indicate correct predictions; off-diagonal entries reveal confusion patterns between visually similar fracture types and the background class.

The difference between mAP50 and mAP50-95 shows how accurate the bounding-box placement is. The predicted boxes for classes with a small gap are tight and well-aligned (compression-crush: 0.844 $\rightarrow$ 0.773). The model finds the fracture region but produces looser bounds for classes with a bigger gap (avulsion: 0.338 $\rightarrow$ 0.278). For a screening tool aimed at radiologists, this trade-off is acceptable.

V. DISCUSSION

A. Why U-net Performed Well

Two factors can explain the low-test loss and high segmentation accuracy achieved by U-Net. First, FracAtlas uses expert-annotated pixel masks as ground truth, providing the model with a clear, superior supervisory signal. Second, the encoder-decoder architecture with skip connections is ideal for fracture detection: the decoder uses skip-connected high-resolution features to recover fine spatial detail (the precise line of the fracture), while the encoder captures global context (which bone, what region). Even more stable training is provided under extreme class imbalance when BCE and dice loss are combined.

B. Why YOLOv8 Underperformed in Rare Classes

YOLOv8's overall mAP50 of 0.364 indicates how difficult it is to detect multi-class fractures across 17 categories with a small dataset. There are two main reasons why strong classes (closed-simple at 0.891 and compression-crush at 0.844) are different from weak classes (longitudinal at 0.081 and oblique at 0.135). The first is that each category that isn't doing well has only 7 to 18 training examples. The second is visual inconsistency, meaning that when fractures are nondisplaced, they may appear as overlapping projections or as normal bone

structure. The 0.42 confidence level for the greenstick example shows calibrated uncertainty. This means that an overconfident wrong prediction could be worse than a tentative right one. A low confidence score, on the other hand, tells the radiologist that the case needs more attention.

C. Clinical Deployment Considerations

There are two important considerations when using Fair-Fracture in a real clinical setting. The first is how well they work with computers: both models are small enough to run on cheap hardware, and because they run at the same time, the slower of the two models limits the system's wall-clock delay instead of their sum. The second is how easy it is to understand: U-Net's predicted mask shows radiologists how confident the model is across the entire image. The YOLOv8 model shows how unsure each prediction is. Neither type of output replaces expert judgment; instead, they support it.

VI. LIMITATIONS AND FUTURE WORK

A. Limitations

It's important to admit that there are some problems with the current system. The most important factors are the dataset size and class imbalance. The Roboflow Bone Fracture dataset contains more than 2,000 photos, but they are not evenly distributed across 17 categories. This means that there are fewer than 20 training samples for each type of rare fracture. Another problem is that it only takes one view of the input: To check three-dimensional alignment, doctors usually use two orthogonal X-ray scans to diagnose fractures. Our pipeline, on the other hand, only works with one image at a time. A third problem is the lack of clinical validation. The third drawback is a lack of clinical validation. We have not compared the system against radiologist interpretations on a sequence of consecutive emergency department X-rays but have tested it on held-out test images from the same source distributions as the training data.

Finally, while the framework is called Fair-Fracture, the present evaluation has not yet performed a systematic fairness audit. As these datasets are geographically limited, they may contain biases specific to the sites. Future research should explicitly focus on measuring performance differences across age, sex, anatomical site, and hardware vendor [11].

B. Future Work

The easiest step forward would be to expand the dataset. To improve the results of YOLOv8 on the classes that are currently underrepresented, at least 200–500 images should be collected for each type of fracture. Multi-view fusion is another priority, where the pipeline takes paired Anteroposterior (the X-ray beam enters from the front (anterior) of the body and exits the back (posterior)) and lateral (the X-ray beam goes through the body side-to-side, giving you a "side view" of the bone.) X-ray inputs to better align with standard clinical workflow [10]. A third priority is a systematic fairness audit, with stratification of performance by acquisition of hardware and patient demographics [11]. Longer-term

objectives include extending volumetric (CT) data using 3D U-Net variants [3] for complex fracture types where 2D X-ray imaging is fundamentally limited and conducting a prospective clinical study comparing system performance with radiologist interpretation on consecutive emergency department X-rays.

VII. CONCLUSION

This paper proposes Fair-Fracture, a deep learning framework that uses a YOLOv8 model for multi-class detection and a U-Net pixel-level segmentation to provide a complete diagnosis of bone fractures from a single X-ray. U-net branch got 99.26%-pixel accuracy on the FracAtlas test set. For 17 fracture classes, YOLOv8 had a mean average precision (mAP50) of 0.364. It did well on closed-simple (0.891) and compression-crush (0.844) fractures. To give radiologists three different pieces of evidence to check against each other, the system runs two models simultaneously with fused outputs, performing fracture classification, bounding-box localization, and pixel-level extent. The multi-class dataset will be enlarged, multi-view inputs will be incorporated, systematic fairness audits will be executed, and prospective clinical validation will be enhanced.

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